

## **ECA5612 –Advanced Wireless Communication Systems 5G / 6G Communication Systems.**

### **Ex.No:1: Simulation of FDMA in MATLAB Date.**

#### **Objective 1:**

To simulate an FDMA system by allocating individual frequency channels to multiple users and determine the channel bandwidth assigned to each user.

#### **Program:**

```
clc;
clear;
close all;
fs = 1000;
t = 0:1/fs:1;

TBW = 600;

Users = 3;

CBW = TBW/Users;

fc = [100 300 500]; % Carrier frequencies m
m = sin(2*pi*10*t); % Message signal

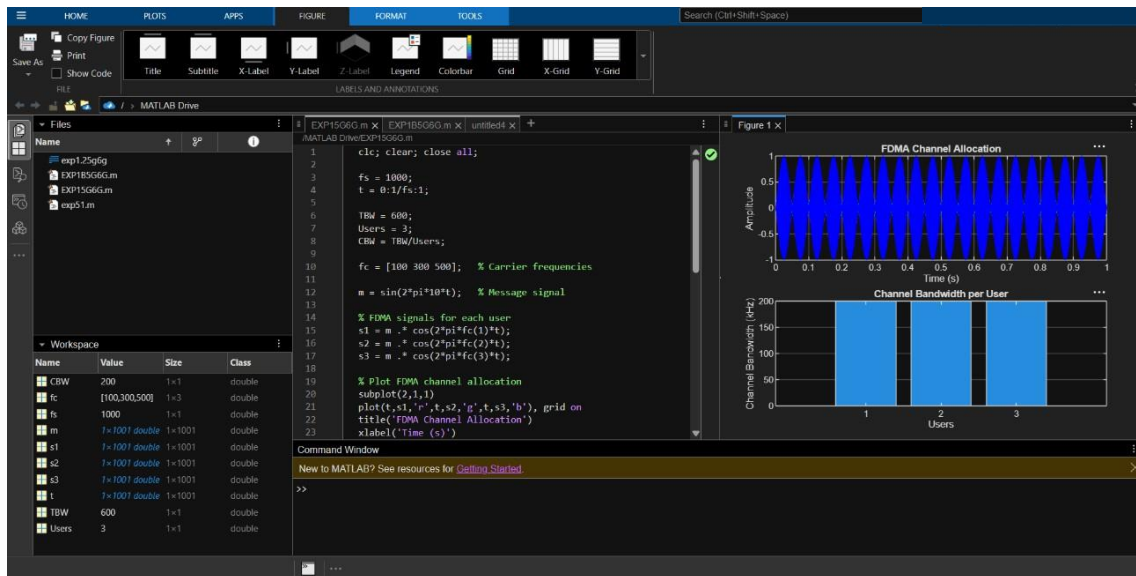
% FDMA signals for each user s1 =
m .* cos(2*pi*fc(1)*t);
s2 = m .* cos(2*pi*fc(2)*t); s3
= m .* cos(2*pi*fc(3)*t);

% Plot FDMA channel allocation
subplot(2,1,1)
plot(t,s1,'r',t,s2,'g',t,s3,'b'), grid on
title('FDMA Channel Allocation')
xlabel('Time (s)') ylabel('Amplitude')

% Plot channel bandwidth per user subplot(2,1,2)
```

```
bar(1:Users, CBW*ones(1,Users)), grid on
xlabel('Users')
ylabel('Channel Bandwidth (kHz)') title('Channel
Bandwidth per User')
```

## OUTPUT:



## Objective 2:

To calculate the data rate achieved by each user in the FDMA system based on the allocated channel bandwidth.

## Program:

```
clc;
clear;
close all;

BW = 200; % Channel Bandwidth (kHz)
Users = 3;
M = 2; % BPSK Modulation

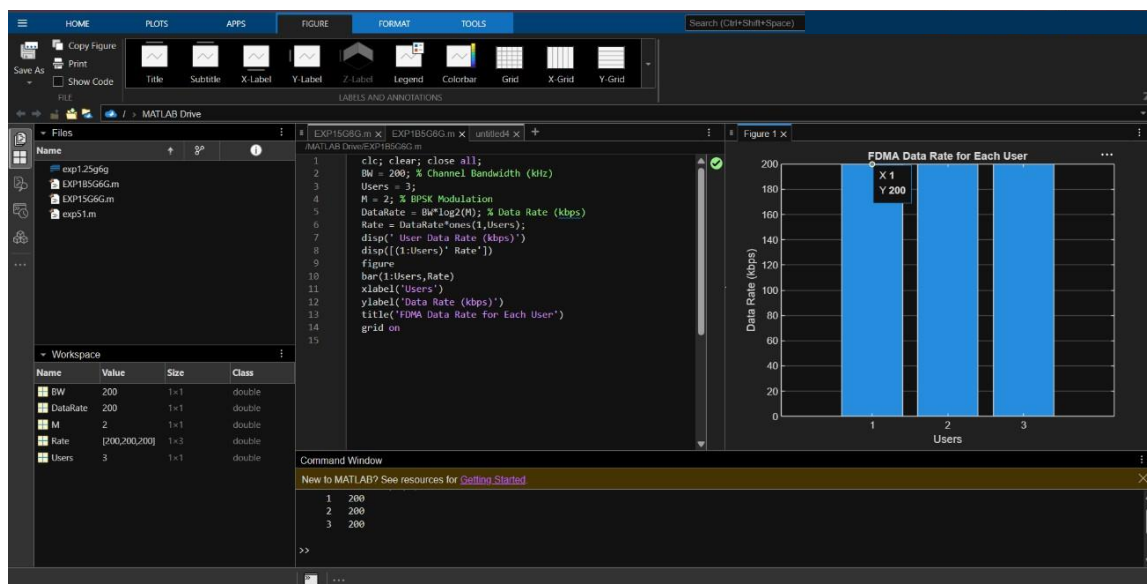
DataRate = BW*log2(M); % Data Rate (kbps)
Rate = DataRate*ones(1,Users);
```

```

disp(' User Data Rate (kbps)')
disp([(1:Users)' Rate'])
figure
bar(1:Users,Rate)
xlabel('Users')
ylabel('Data Rate (kbps)')
title('FDMA Data Rate for Each User') grid
on

```

## OUTPUT:



## Objective 3:

To evaluate the communication quality of the FDMA system by calculating the Signal-to

Noise Ratio (SNR) for different users.

## Program:

```

clc;

clear;

close all;

Ps = 1; % Signal Power (W)

Pn = [0.1 0.05 0.01]; % Noise Power (W)

Users = 1:3;

```

```

SNR = 10*log10(Ps./Pn); % SNR in dB

disp('User Noise Power(W) SNR(dB)')

disp([Users' Pn' SNR'])

figure bar(Users,SNR)

xlabel('Users')

ylabel('SNR (dB)')

title('Signal-to-Noise Ratio of FDMA Users') grid

on

```

## OUTPUT:

